

Extra Task: 7 Segment Display

```
/*
 * File:  main.c
 * Author:  sagar
 *
 * Created on 22 May, 2025, 12:12 PM
 */

#include <xc.h>
#include <stdint.h>

// CONFIG
#pragma config FOSC = EXTRC    // Oscillator Selection bits (RC oscillator)
#pragma config WDTE = OFF      // Watchdog Timer Enable bit (WDT enabled)
#pragma config PWRT = OFF      // Power-up Timer Enable bit (PWRT disabled)
#pragma config BOREN = OFF     // Brown-out Reset Enable bit (BOR enabled)
#pragma config LVP = OFF       // Low-Voltage (Single-Supply) In-Circuit Serial Programming
Enable bit (RB3/PGM pin has PGM function; low-voltage programming enabled)
#pragma config CPD = OFF       // Data EEPROM Memory Code Protection bit (Data EEPROM
code protection off)
#pragma config WRT = OFF       // Flash Program Memory Write Enable bits (Write protection off;
all program memory may be written to by EECON control)
#pragma config CP = OFF        // Flash Program Memory Code Protection bit (Code protection off)

#define _XTAL_FREQ 20000000

char pattern_1[10] = {0xF9,0xA4,0xB0,0x99,0x92,0x82,0xF8,0x00,0x10,0x40}; // Common
cathode
char pattern_2[]={0x06,0x5B,0x4F,0x66,0x6D,0x7D,0x07,0xFF,0xEF,0xBF}; // Common Anode

void main(void)
{
    TRISD = 0x00;
    TRISB = 0x00;

    PORTB = 0x00;
    PORTD = 0x00;

    while(1)
    {
        for(int i = 0; i < 10; i++)
        {
            PORTD = binary_pattern_2[i];
            PORTB = binary_pattern_1[i];
            __delay_ms(1000);
        }
    }
    return;
}
```